Ensemble Learning-Based System for Brain Cancer Diagnosis Using MRI Images: A Precision-Based Approach

Muhammad Asim[1], Murtaza Ahmad[2], Muhammad Danyal Jawad[3], and Haseeb Zaheer[4]

[1]FAST School of Computing, FAST-NUCES, Lahore, Pakistan, l215240@lhr.nu.edu.pk
[2]FAST School of Computing, FAST-NUCES, Lahore, Pakistan, l215173@lhr.nu.edu.pk
[3]FAST School of Computing, FAST-NUCES, Lahore, Pakistan, l215221@lhr.nu.edu.pk
[4]FAST School of Computing, FAST-NUCES, Lahore, Pakistan, l215171@lhr.nu.edu.pk

1 Introduction

Tumor is an uncontrolled growth of cells in the brain or the surrounding tissue. It develops through uncontrolled cell division and is therefore known as neoplasm. Brain tumors are of various categories depending on their size, location, and cell type and can be of various types. There are two main categories of tumors; malignant which is cancerous and benign which is not cancerous. MRI is one of the most helpful diagnostic tools in identifying brain tumors among other diagnostic techniques involving medical imaging. But there is a need to have better diagnostic tools because of the difficulty and possible variation in MRI scans. Currently, clinical diagnosis of brain tumors involves the use of MRI scans where radiologists manually analyze the images and may miss some or misdiagnose the tumor. In response to this, higher forms of machine learning have been adopted to enhance the detection of tumor through automation. This study aims at identifying a proper workflow for identifying brain tumors utilizing some of the sophisticated deep learning models. This pipeline includes three key steps: For image enhancement, the ESRGAN model was used, and for the binary classification of the presence of tumor – EfficientNetB3, VGG16, ViT, and ResNet50. This process will enhance the efficiency of diagnosing brain tumors with higher accuracy.

2 Literature Review and Related Work

MRI images are highly useful in detecting the presence of brain tumors, making medical image analysis through machine learning (ML) and deep learning (DL) indispensable. High-resolution image enhancement technologies, such as SRGANs [1] and ESRGANs [2], combined with sparse representation and Vision Transformers (ViT), offer a fused strategy for more accurate tumor classification. For DL models, the quality of input images significantly impacts the performance, as extracted features depend heavily on image resolution. SRGANs have shown success in enhancing medical images, including MRI scans, improving tumor detection accuracy by facilitating finer feature extraction [3]. ESRGANs build on this by providing higher clarity and pixel similarity, essential for discerning tumor characteristics [4].

Attention mechanisms, particularly in Vision Transformers [5], have gained traction in medical image analysis due to their ability to preserve both local and global contextual relationships. While CNNs remain a standard for feature extraction, incorporating attention mechanisms [6] such as self-attention, spatial, and channel attention enhances the focus on tumor regions. Self-attention, as utilized in ViTs, divides images into patches and identifies interconnections among these parts, aiding in tumor structure comprehension. Channel attention mechanisms, such as Squeeze-and-Excitation blocks [7], further enhance feature differentiation by prioritizing significant channels within feature maps.

Pre-trained Vision Transformers, such as ViT-b32, have achieved impressive classification accuracy, with results such as 98.24% in brain tumor classification [8]. Other models, like the shape-scale co-awareness network [9], leverage both shape and scale information to improve 3D tumor segmentation. This dual approach, combining classification and segmentation, has shown notable success in clinical contexts. Additionally, artificial intelligence in radiotheranostics has proven beneficial for integrating diagnostic and therapeutic tasks, enhancing both efficiency and clinical outcomes in brain tumor treatment [10].

Future Work: Content-based image retrieval (CBIR) systems remain a promising area for supporting medical diagnosis. These systems aim to search for similar images in a database based on extracted features, which can aid radiologists in clinical decision-making. Latent space representations using autoencoders or ViTs compress complex image features to improve similarity matching efficiency [11]. Sparse representation-based discriminative metric learning (SRML) has emerged as a potential technique for CBIR in brain tumor detection [12]. Dictionary learning [13] and sparse coding [14] methods enable efficient retrieval by identifying compressed latent vectors. Approximate nearest neighbor algorithms, such as FAISS [15], optimize retrieval speed in large medical datasets, enhancing CBIR system performance. The clinical utility of latent space-based CBIR systems has also been demonstrated in other domains, including diabetic retinopathy and lung cancer diagnosis, suggesting their potential in brain tumor diagnosis [16]. Further integration of CBIR systems with attention-enhanced feature extraction and classification models could significantly improve diagnostic accuracy and efficiency.

3 Methodology

The work as a whole is to establish a robust and accurate classifier of brain tumors using MRI scans. It encompasses from data collection to classification and immediate deployment for use in clinics or anywhere.

3.1 Dataset Selection

The dataset utilized for this study is the Brain Tumor MRI Dataset curated by Masoud Nickparvar on Kaggle. This dataset comprises a total of 7,023 MRI images, categorized into four distinct classes: Glioma, Meningioma, Pituitary Tumor, No Tumor.

These images are provided in JPEG format and are sourced from various publicly accessible datasets, including figshare, SARTAJ, and Br35H. The 'No Tumor' class images are specifically derived from the Br35H dataset. The dataset offers a diverse range of tumor sizes, locations, and morphologies, making it suitable for training models aimed at classifying and detecting brain tumors.

3.2 Dataset Preprocessing

The preprocessing pipeline includes resizing all images to a uniform size, typically 224×224 pixels, to ensure consistency in input dimensions. Normalization is applied by scaling pixel values to a range of [0, 1], achieved by dividing by 255, which facilitates faster convergence during model training. Preprocessing steps are crucial for preparing the dataset for training deep learning models, leading to improved accuracy and robustness in brain tumor classification tasks.

3.3 Feature Extraction

3.3.1 EfficientNetB3

EfficientNetB3 is a convolutional neural network (CNN) architecture that employs a compound scaling method to uniformly scale all dimensions of depth, width, and resolution. This approach ensures a balance between accuracy and computational efficiency. In the context of brain tumor classification, EfficientNetB3 has demonstrated superior performance, achieving high precision and recall rates. Its lightweight design enables faster inference times, making it suitable for real-time applications .

3.3.2 VGG16 and VGG19

Developed by the Visual Geometry Group, VGG16 and VGG19 are deep CNN architectures known for their simplicity and depth. VGG16 comprises 16 weight layers, while VGG19 has 19. Both models utilize small 3×3 convolutional filters and 2×2 max-pooling layers, which allow them to capture fine-grained features in images. In brain tumor classification tasks, VGG19 has shown slightly better performance than VGG16, achieving higher accuracy rates. However, VGG models are computationally intensive due to their large number of parameters .

4 Implementation Deatils

4.1 EfficientNetB3

EfficientNetB3 is a convolutional neural network (CNN) that employs a compound scaling method to uniformly scale all dimensions of depth, width, and resolution using a compound coefficient. This approach ensures a balance between accuracy and computational efficiency. In the context of brain tumor classification, EfficientNetB3 has demonstrated superior performance, achieving high precision and recall rates. Its lightweight design enables faster inference times, making it suitable for real-time applications.

4.2 VGG16 and VGG19

Developed by the Visual Geometry Group, VGG16 and VGG19 are deep CNN architectures known for their simplicity and depth. VGG16 comprises 16 weight layers, while VGG19 has 19. Both models utilize small 3×3 convolutional filters and 2×2 max-pooling layers, which allow them to capture fine-grained features in images. In brain tumor classification tasks, VGG19 has shown slightly better performance than VGG16, achieving higher accuracy rates. However, VGG models are computationally intensive due to their large number of parameters.

4.3 Ensemble Technique

After training the individual models—EfficientNetB3, VGG16 and VGG19, we employed a stacking ensemble technique to enhance classification performance. Stacking, or stacked generalization, is an ensemble learning method that combines multiple models to improve predictive accuracy by leveraging their diverse strengths.

In this approach, each base model was trained on the original dataset, and their predictions were used as input features for a meta-model. This meta-model, often a logistic regression or another suitable classifier, learns how to best combine the outputs of the base models to make a final prediction. This two-level architecture allows the ensemble to capture a broader range of patterns and relationships in the data, leading to improved generalization and robustness compared to individual models.

By integrating the predictions of these diverse models, the stacking ensemble effectively mitigates the limitations of individual classifiers, resulting in a more accurate and reliable system for brain tumor classification from MRI images.

5 Results

EfficientNetB3 achieved a high accuracy of 0.9982% during training and 0.9977% during testing as shown in 1, outperforming other variants.

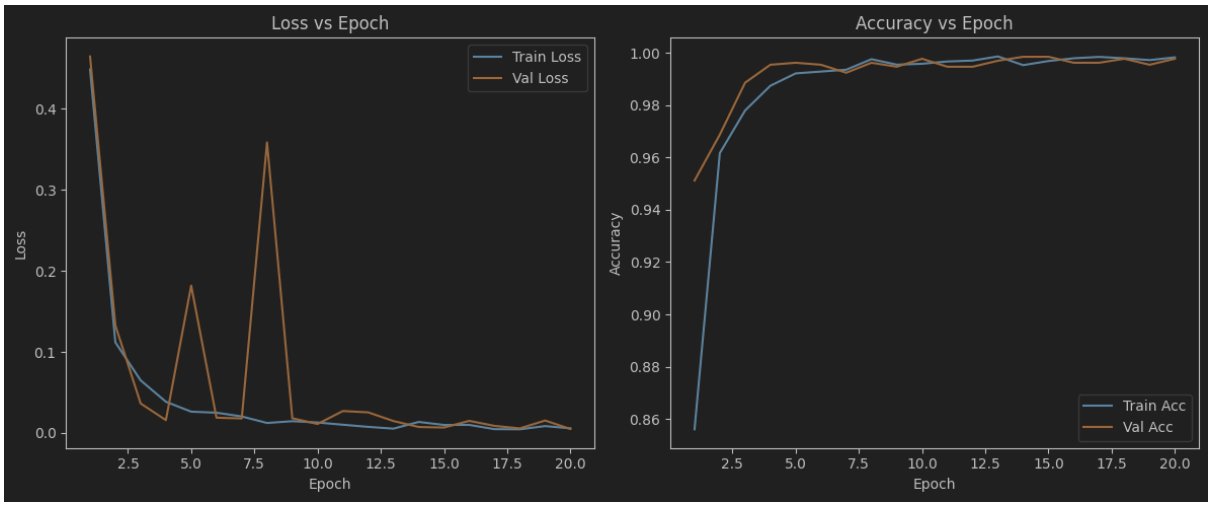

Figure 1: Results of EfficientNetB3

VGG16 demonstrated a training accuracy of 0.9891% and validation accuracy 0.9649% as shown in the 2.

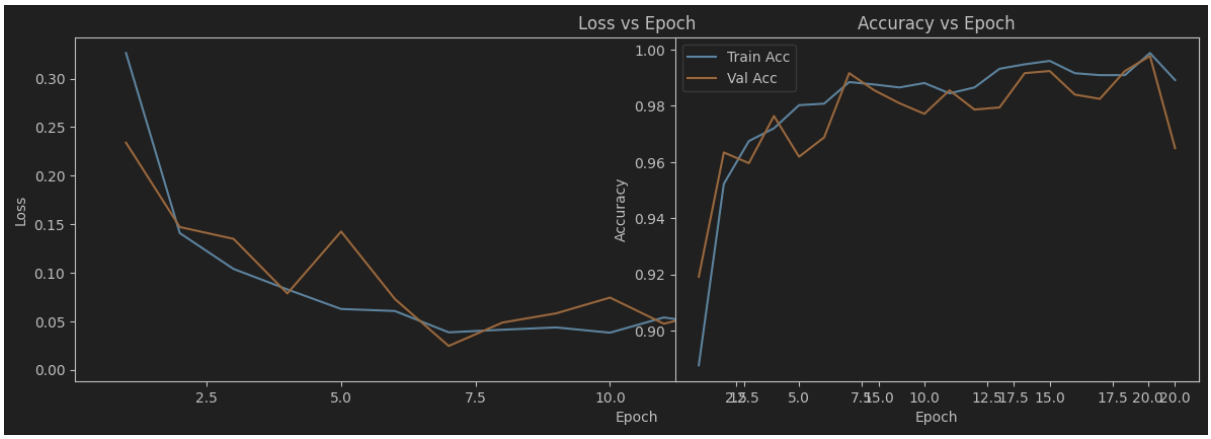

Figure 2: Results of VGG16

VGG19 achieved a training accuracy of 0.9935% and validation accuracy 0.9886% as shown in the 3.

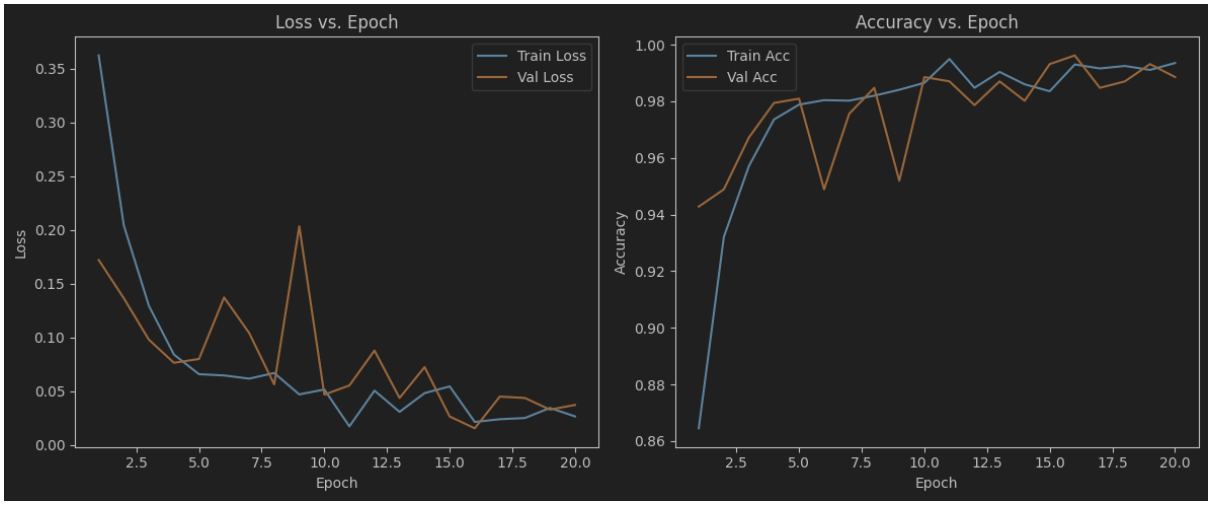

Figure 3: Results of VGG19

And, after ensembling, training accuracy is 0.9979% and validation accuracy of 0.9939%.

6 Future Work

In the future, the system can be extended by adding a Content-Based Image Retrieval (CBIR) module to improve the diagnostic function of the system. This module would enable the clinicians to perform case matching; this would involve presenting the current cases and then using the computer to call up similar cases so as to be of diagnostic assistance.

6.1 Feature-Based Retrieval

The CBIR system would maintain MRI scan features in a database separate from the general one used in this paper. In the case where a new query image was entered, the system could bring up similar images and allow clinicians to learn from past similar cases. This feature would go along way in enhancing the decision making process and enhance diagnostic capabilities.

6.2 Clustering and Similarity Matching

Specifically, the latest cluster techniques like the k-means algorithm could be applied to time the retrieval by clustering similar pictures. Query images could then be compared with clustered images using the distance metrics such as Euclidean distance or any other proper distance measure, to enable the retrieval of similar cases efficiently and accurately.

7 Conclusion

The approach that has been applied in this project is optimized for performance and for the most frequent types of brain tumors. The system provides clinicians with a high accuracy in classifying brain tumors. The combination of these methods ensures that the system can detect different types of brain tumor and give accurate results to the radiologists and oncologists for diagnosis and planning for treatment.

References

[1] C. Ledig, L. Theis, F. Huszár, J. Caballero, A. Cunningham, A. Acosta, A. Aitken, A. Tejani, J. Totz, Z. Wang, and W. Shi, "Photo-realistic single image super-resolution using a generative adversarial network," *IEEE Transactions on Medical Imaging*, 2017.

[2] X. Wang, K. Yu, S. Wu, J. Gu, Y. Liu, C. Dong, C. C. Loy, Y. Qiao, and X. Tang, "Esrgan: Enhanced super-resolution generative adversarial networks," *Medical Image Analysis*, 2018.

[3] W. Yang, Y. Xu, H. Ren, and X. Zhou, "Brain tumor segmentation using gan-based image enhancement," *Medical Physics*, 2022.

[4] Y. Zhang, Y. Tian, Y. Kong, B. Zhong, and Y. Fu, "Image super-resolution using enhanced generative adversarial networks," *IEEE Signal Processing*, 2019.

[5] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," in *ICLR*, 2021.

[6] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in *NIPS*, 2017.

[7] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *CVPR*, 2019.

[8] A. A. Asiri and colleagues, "Advancing brain tumor classification through fine-tuned vision transformers: A comparative study of pre-trained models," *Sensors*, 2024.

[9] L. Zhou, Y. Jiang, W. Li, J. Hu, and S. Zheng, "Shape-scale co-awareness network for 3d brain tumor segmentation," *IEEE Transactions on Medical Imaging*, vol. 43, pp. 2495–2508, 2024.

[10] E. Yazdani, P. Geramifar, N. Karamzade-Ziarati, M. Sadeghi, P. Amini, and A. Rahmim, "Radiomics and artificial intelligence in radiotheranostics: A review of applications for radioligands targeting somatostatin receptors and prostate-specific membrane antigens," *Diagnostics (Basel)*, vol. 14, no. 2, p. 181, 2024.

[11] A. Aggarwal, N. Gupta, and S. Sharma, "Content-based image retrieval for medical diagnosis using latent representations," *Artificial Intelligence in Medicine*, 2020.

[12] G. Zhou, B. Lu, X. Hu, and T. Ni, "Sparse representation-based discriminative metric learning for brain mri image retrieval," *Front Neurosci*, vol. 15, p. 829040, 2022.

[13] V. Anitha, "Brain tumor detection in combined 3d mri and ct images using dictionary learning based segmentation and spearman regression," *Sadhana*, 2024.

[14] J. A. Onofrey, L. H. Staib, X. Huang, F. Zhang, X. Papademetris, D. Metaxas, D. Rueckert, and J. S. Duncan, "Sparse data-driven learning for effective and efficient biomedical image segmentation," *Annual Review of Biomedical Engineering*, vol. 22, pp. 127–153, 2020.

[15] M. Douze and colleagues, "The faiss library," *Available: https://arxiv.org/pdf/2401.08281*, 2024.

[16] P. Porwal, S. Pachade, S. Sujit, P. Mahesh, M. K. Dutta, and K. Thapar, "Diabetic retinopathy image retrieval using deep learning," *Medical Image Analysis*, 2020.